Xiyang Wu

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Education

University of Maryland — College Park, MD
Ph.D. in Electrical and Computer Engineering, GAMMA Laboratory — *Aug. 2021 – Dec. 2026 (Expected)*
Advisor: Dinesh Manocha GPA: 3.83/4.00

Georgia Institute of Technology — Atlanta, GA
M.S. in Electrical and Computer Engineering, CORE Robotics Laboratory — *Aug. 2019 – May. 2021*
Advisor: Matthew Gombolay GPA: 4.00/4.00

Tianjin University — Tianjin, China
B.Eng. in Measuring and Controlling Technologies and Instruments (Honors Class) — *Sep. 2015 – Jul. 2019*
Advisor: Xiaodong Zhang GPA: 3.85/4.00

Publication

(* indicates equal contributions)

1. **Xiyang Wu**, Zongxia Li, Guangyao Shi, Alexander Duffy, Tyler Marques, Matthew Lyle Olson, Tianyi Zhou, Dinesh Manocha. Co-Evolving LLM Decision and Skill Bank Agents for Long-Horizon Tasks. *arXiv:2604.20987, 2026*, Link, Code, Project Page, Model.
2. **Xiyang Wu**, Guangyao Shi, Qingzi Wang, Zongxia Li, Amrit Singh Bedi, Dinesh Manocha. SABER: A Stealthy Agentic Black-Box Attack Framework for Vision-Language-Action Models. *arXiv:2603.24935, 2026*, Link.
3. **Xiyang Wu**, Zongxia Li, Jihui Jin, Guangyao Shi, Nilotpal Sinha, Gouthaman KV, Vishnu Raj, Fan Du, Dinesh Manocha. MASS: Motion-Aware Spatial–temporal Grounding for Physics Reasoning and Comprehension in Vision-Language Models. *arXiv:2511.18373, CVPR 2026.* Link,
4. **Xiyang Wu***, Zongxia Li*, Yubin Qin, Guangyao Shi, Hongyang Du, Dinesh Manocha, Tianyi Zhou, Jordan Lee Boyd-Graber. VideoHallu: Enhancing Multimodal Synthetic Video Understanding via Reinforcement Curriculum Learning. *arXiv:2505.01481, NeurIPS 2025.* Link, Code, Project Page, Dataset.
5. **Xiyang Wu**, Souradip Chakraborty, Ruiqi Xian, Jing Liang, Tianrui Guan, Fuxiao Liu, Brian Sadler, Dinesh Manocha, Amrit Singh Bedi. Highlighting the Safety Concerns of Deploying LLMs/VLMs in Robotics. *arXiv:2402.10340, IROS 2025.* Link, Code, Project Page.
6. **Xiyang Wu***, Zongxia Li*, Hongyang Du, Huy Nghiem, Guangyao Shi. Benchmark evaluations, applications, and challenges of large vision language models: A survey. *arXiv:2501.02189, CVPR 2025 Workshop.* Link.
7. **Xiyang Wu***, Tianrui Guan*, Dianqi Li, Shuaiyi Huang, Xiaoyu Liu, Xijun Wang, Ruiqi Xian, Abhinav Shrivastava, Furong Huang, Jordan Lee Boyd-Graber, Tianyi Zhou, Dinesh Manocha. AUTOHALLUSION: Automatic Generation of Hallucination Benchmarks for Vision-Language Models. *arXiv:2406.10900, EMNLP 2024.* Link, Project Page.
8. **Xiyang Wu***, Chak Lam Shek*, Wesley A. Suttle, Carl Busart, Erin Zaroukian, Dinesh Manocha, Pratap Tokekar, Amrit Singh Bedi. LANCAR: Leveraging Language for Context-Aware Robot Locomotion in Unstructured Environments. *IROS 2024.* Link, Project Page.
9. **Xiyang Wu**, Rohan Chandra, Tianrui Guan, Amrit Singh Bedi, Dinesh Manocha. Intent-Aware Planning in Heterogeneous Traffic via Distributed Multi-Agent Reinforcement Learning. *CoRL 2023 (Oral)*, *IROS 2023 MRS Workshop (Best Paper Award).* Link, Code.
10. Qingzi Wang, **Xiyang Wu**, Guangyao Shi, Dianwei Chen, Xianfeng Yang, Dinesh Manocha Act on What You See: Unlocking Safe Social Navigation in Vision-Language-Action Models. *Under Submission, 2026.*
11. Zongxia Li, Hongyang Du, Chengsong Huang, **Xiyang Wu**, Lantao Yu, Yicheng He, Jing Xie, Xiaomin Wu, Zhichao Liu, Jiarui Zhang, Fuxiao Liu MM-Zero: Self-evolving multi-model vision language models from zero data. *arXiv:2603.09206, 2026.* Link.
12. Jingxi Chen, Zongxia Li, Zhichao Liu, Guangyao Shi, **Xiyang Wu**, Fuxiao Liu, Cornelia Fermuller, Brandon Y. Feng, Yiannis Aloimonos. First Frame Is the Place to Go for Video Content Customization. *arXiv:2511.15700, CVPR 2026.* Link, Code, Project Page.

13. Haoyue Liu, **Xiyang Wu**, Ning Yan, Zexiao Li, Xiaodong Zhang. A novel image registration-based dynamic photometric stereo method for online defect detection in aluminum alloy castings. *Digital Signal Processing*, 2023.
14. Hossein Khayami, Sungjin Hwang, **Xiyang Wu**, Eshed Ohn-Bar, David E. Conroy, Amanda Lazar, Eun Kyoung Choe, Hernisa Kacorri. Older Adults Personalizing Wearable Foundation Models: Feasibility and Challenges. *Under Submission*, *IMWUT*, 2025.
15. Ming Li, Chenguang Wang, Yijun Liang, Xiyao Wang, Yuhang Zhou, **Xiyang Wu**, Yuqing Zhang, Ruiyi Zhang, Tianyi Zhou. CaughtCheating: Is Your MLLM a Good Cheating Detective? *arXiv:2507.00045.* Link.
16. Zongxia Li, Yapei Chang, Yuhang Zhou, **Xiyang Wu**, Zichao Liang, Yoo Yeon Sung, Jordan Lee Boyd-Graber. Semantically-Aware Rewards for Open-Ended R1 Training in Free-Form Generation. *arXiv:2506.15068.* Link, Code.
17. Ruiqi Xian, **Xiyang Wu**, Tianrui Guan, Xijun Wang, Boqing Gong, Dinesh Manocha. SOAR: Self-supervision Optimized UAV Action Recognition with Efficient Object-Aware Pretraining. *arXiv:2409.18300, Submitted to IEEE RA-L.* Link.
18. Tianrui Guan*, Ruiqi Xian*, Xijun Wang, **Xiyang Wu**, Mohamed Elnoor, Daeun Song, Dinesh Manocha. AGL-NET: Aerial-Ground Cross-Modal Global Localization with Varying Scales. *IROS 2024.* Link.
19. Tianrui Guan*, Fuxiao Liu*, **Xiyang Wu**, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, Dinesh Manocha, Tianyi Zhou. HallusionBench: An Advanced Diagnostic Suite for Entangled Language Hallucination and Visual Illusion in Large Vision-Language Models. *CVPR 2024.* Link, Code.
20. Esmaeil Seraj, **Xiyang Wu**, Matthew Gombolay. FireCommander: An Interactive, Probabilistic Multi-agent Environment for Joint Perception-Action Tasks. *arXiv:2011.00165.* Link, Code.

Working Experience

NEC Laboratories America, Inc. — Princeton, NJ
Research Intern — *Mentor: Deep Patel, Iain Melvin, Martin Renqiang Min* — *Jun. 2026 – Aug. 2026*

- **Streaming Video LLM with Infinite Memory.** Developing a streaming Video-LLM framework with external long-term memory for infinite-context video understanding, focusing on efficient stream processing, multi-modal memory storage nadretrieval, and agentic multi-hop temporal reasoning over day-scale egocentric videos.

Dolby Laboratories, Inc. — Atlanta, GA
Research Intern — *Mentor: Jihui Jin, Gouthaman KV, Vishnu Raj, Nilotpal Sinha* — *May. 2025 – Aug. 2025*

- **Mitigating Hallucination in Synthetic Video Question-answering.** Enhanced foundation model synthetic video understanding abilities with spatial and motion-awareness via depth estimation, visual grounding, and motion tracking; applied instruction tuning and curriculum reinforcement fine-tuning on real and synthetic video datasets.

Research Experience

GAMMA Laboratory, University of Maryland — College Park, MD
Research Assistant — *Advisor: Dinesh Manocha* — *Sep. 2022 – Now*

- **Evolving Multi-modal Agentic Memory for Long-Horizon Decision Making and Long-Context Understanding** (In progress). Developing agentic memory systems that continuously organize, update, and retrieve reusable experiences/skills to support long-horizon planning, context-aware reasoning, and robust decision making over extended interaction histories.
- **Physics-Aware Synthetic Video Understanding and Generation** (In progress). Leveraging multi-modal foundation models to improve synthetic video understanding and evaluation, enabling physics-aware, higher-quality video generation.
- **Automated Adversarial Agents for Probing Embodied AI Vulnerabilities** (In progress). Developing multi-modal foundation model agents to adversarially attack reasoning in Vision-Language -Action (VLA) models.
- **Hallucinations in Multi-modal LLMs**. Investigating methods to detect and mitigate hallucinations in multi-modal foundation modes' reasoning and question answering.

- **Intent-aware Autonomous Driving**. Developed a distributed multi-agent RL algorithm that jointly predicts trajectories and intents in dense traffic, integrating high-level behavioral incentives for decision-making and low-level incentives for motion planning.

Cognitive Optimization and Relational (CORE) Robotics Laboratory Atlanta, GA
Research Assistant *Advisor: Matthew Gombolay* *Jan. 2020 – Dec. 2020*

- **FireCommander: Multi-agent Wildfire Pruning System.** Implemented state-of-the-art reinforcement learning methods in a custom simulation for multi-agent firefighting tasks.

Laboratory of Micronano Manufacturing Technology Tianjin, China
Research Assistant *Advisor: Xiaodong Zhang* *Sep. 2018 – Jul. 2019*

- **Online Scratch Inspection System.** Designed a defect detection system using photometric stereo and advanced image processing techniques.

Skills

- **Programming:** Python, C/C++, MATLAB, Shell, JavaScript
- **Machine Learning:** PyTorch, Deep Learning, Reinforcement Learning, RLHF
- **Foundation Models:** LLMs, Vision–Language Models, Multimodal & Video Understanding, Visual Reasoning, Diffusion Models, World Models, Post-Training
- **Robotics & Embodied AI:** ROS, Robot Navigation & Manipulation, Control, Vision–Language–Action Models, Autonomous Driving
- **Research & Collaboration:** Scientific Writing, Teaching, Mentorship, Leadership

Teaching Experience

Graduate Teaching Assistant *University of Maryland*

ENEB 354: Discrete Mathematics for Information Technology *Fall 2024*

ENEE 664: Optimal Control *Spring 2023*

ENEE 245: Digital Circuits and Systems Laboratory *Spring 2023, Spring 2025*

ENEE 303: Analog and Digital Electronics *Fall 2022*

ENEE 307: Electronic Circuits Design Laboratory *Spring 2022*

ENEE 322: Signal and System Theory *Fall 2021*

Honor & Awards

- Best Paper Award, IROS 2023 MRS Workshop
- Merit Student Award in Tianjin University, 2018
- Samsung Scholarship, 2017
- Secondary Scholarship in Hexagon Innovation Laboratory in Tianjin University, 2016
- National Secondary Award in the 10th iCAN International Contest of Innovation, 2016

Academic Service

- **Journal Reviewer:** *IEEE Access*, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *Journal of Medical Internet Research (JMIR)*, *IEEE Robotics and Automation Letters (RA-L)*
- **Conference Reviewer:** *NeurIPS*, *CVPR*, *CoRL*, *ICRA*, *IROS*, *EMNLP*, *NAACL*, *ACL*
- **Program Committee Member:** CoCoMARL Workshop at RLC
- **Graduate Application Committee:** University of Maryland
- **TA Training and Development (TATD) Fellow:** 2025–2026, Department of lectrical and Computer Engineering, University of Maryland